

AI Shop Sync: A Mobile-First OCR-Driven Price Comparison Framework with Cross-Platform Aggregation and In-Place Machine Learning

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Abstract—The rapid expansion of online grocery and hyper-local delivery platforms has intensified the need for intelligent tools capable of comparing real-time prices across multiple retailers. Although price comparison applications exist, most rely on barcode scans or manual text search, both of which remain inadequate for the Indian FMCG ecosystem where packaging varies greatly and barcodes are inconsistently printed. This paper presents *AI Shop Sync*, a mobile-first, OCR-enhanced price comparison framework that integrates on-device text recognition, robust linguistic normalization, multi-source web aggregation, and a deployed machine-learning cost optimization engine. The system is engineered for real-world retail environments where lighting, background noise, and inconsistent packaging introduce significant noise to text extraction.

AI Shop Sync employs Google ML Kit for low-latency on-device OCR to reduce server load, preserve user privacy, and ensure sub-200ms recognition speeds. Extracted text is passed to a normalization pipeline that performs noise filtering, unit standardization, semantic variant reconciliation, and canonical mapping to a curated catalog of over 45,000 grocery SKUs. The backend aggregates real-time prices from API-enabled platforms and automated Playwright-based scrapers, producing a harmonized multi-retailer pricing model. A gradient-boosted ML comparator, trained on historical pricing and promotion logs, computes optimal single-vendor or split-cart purchase strategies, delivering up to 22% savings in field validation.

The proposed system was evaluated across 1,200+ scans in 15 retail settings and achieved 98.7% mapping reliability, 99.2% pricing accuracy, and 2.3-second end-to-end response time. The findings demonstrate the feasibility of combining mobile OCR, NLP, and multi-source aggregation into a unified, consumer-ready price intelligence solution. The expanded analysis contributes new practical insights into edge-based OCR deployment, schema alignment across heterogeneous retailers, and ML-powered optimized cart recommendations.

Index Terms—Optical Character Recognition, Price Comparison, Machine Learning, Mobile Computing, Web Scraping, NLP, React Native, Retail Intelligence.

I. INTRODUCTION

The Indian e-grocery and hyperlocal delivery sector has undergone exponential growth over the last decade, driven

by rapid urbanization, digital adoption, and the emergence of instant-delivery platforms. Studies estimate that the sector surpassed USD 8.8 billion in 2024 and is expected to exceed USD 50 billion by 2030. Despite this growth, consumers face persistent inefficiencies when attempting to compare prices across platforms such as Swiggy Instamart, Zepto, Blinkit, BigBasket, and DMart Ready. Price fluctuations, limited transparency in delivery fees, and platform-specific promotional structures create a complex decision-making environment.

Traditional price comparison mechanisms—barcode-based apps, manual keyword search, and standalone comparison websites—fail to address the unique textual variance in Indian FMCG packaging. Many household goods lack standardized barcodes, while packaging often includes decorative fonts, multilingual labeling, and variant descriptors that complicate search-based comparison. Additionally, users frequently scan products under poor lighting or angled conditions, introducing further OCR challenges.

AI Shop Sync addresses these gaps through a mobile-first architecture wherein on-device OCR captures product text directly from the packaging, enabling rapid identification without requiring barcodes or database lookups. The extracted text is processed through a domain-specific normalization pipeline that removes batch codes, harmonizes quantities (e.g., 500g vs 0.5kg), corrects OCR-induced spelling distortions, and maps the cleaned string to a canonical product record.

On the backend, the system integrates real-time retailer prices using both official APIs and controlled web scraping pipelines, ensuring freshness and platform coverage. A trained machine-learning comparator evaluates cost structures by incorporating vendor-specific delivery fees, promotional modifiers, cart thresholds, and item-level price trajectories. The result is a unified, intelligent recommendation that selects the most cost-efficient vendor or split-vendor strategy for the user.

This paper presents an expanded technical analysis of the

system, including OCR performance in real-world environments, normalization accuracy across heterogeneous packaging styles, backend concurrency optimization, caching strategies, and ML training methodology. Additionally, we contextualize our work within existing research on OCR [1]–[4], web data extraction [9], [10], and retail pricing intelligence [13], [15].

II. RELATED WORK

Research on OCR has advanced significantly with deep learning-based architectures. Classical engines such as Tesseract [1] laid foundational work for document OCR, while modern systems like AON [2] and TextOCR [3] demonstrate robust recognition in complex, real-world environments. These advancements have enabled OCR applications in retail product identification and scene-text extraction where fonts, orientations, and lighting vary substantially.

Several studies have contributed surveys and benchmarks for OCR performance. Long et al. [4] provide a comprehensive study of deep learning approaches for text detection and recognition, highlighting improvements in multi-language support and robustness to curved or deformed text. These works inform our choice of Google ML Kit as a mobile-optimized OCR engine due to its low-latency inference and multilingual capabilities.

In the domain of data aggregation and e-commerce analytics, web scraping remains a critical tool. Liu et al. [9] and Zhao and Wu [10] document methods for robust web extraction under dynamic content and anti-bot protection. Studies such as Gupta et al. [11] explore scalable price intelligence systems capable of integrating real-time updates from numerous retail sources.

Related research in retail pricing and recommendations further motivates our ML comparator. Adomavicius and Tuzhilin [13] provide a foundational survey on recommender systems, while Neural Collaborative Filtering [14] and modern price optimization models [15] demonstrate the feasibility of using machine learning for cost prediction and decision support. Li and Jin [16] describe applications of ML for dynamic pricing and discount evaluation in e-commerce.

Mobile computing literature—e.g., Satyanarayanan’s work on edge computing [17] and Shi et al. [18]—supports our choice of performing OCR on-device rather than on the cloud. On-device processing reduces latency, improves privacy, and offloads computational demands from the backend.

Despite extensive research, few systems unify OCR-based product recognition, real-time price comparison, cross-platform aggregation, and ML-based optimization into a single mobile application. AI Shop Sync aims to bridge this gap by combining advances in mobile OCR, NLP, web automation, and retail intelligence into a coherent, user-facing solution.

III. SYSTEM ARCHITECTURE

AI Shop Sync adopts a multi-layered architecture designed to support low-latency OCR, scalable data aggregation, and

machine-learning-based price recommendations. The architecture consists of three primary tiers: the mobile client, backend services, and the integration layer that interfaces with APIs and web extractors. Fig. 1 provides a high-level overview of the system pipeline.

A. A. Mobile Tier

The mobile tier is implemented using React Native, enabling shared code across both Android and iOS. The device acts as the primary computational unit for text extraction, leveraging the Vision Camera library for high-resolution image capture. The mobile client executes:

- On-device OCR using Google ML Kit for low-latency inference.
- Local pre-processing such as blur detection, exposure checks, and frame filtering.
- Secure communication with the backend through JWT-signed requests.
- Local caching of recent scans using SQLite for offline resilience.

The decision to push OCR to the mobile device is aligned with edge computing principles [17], where computation is positioned close to the data source to reduce latency and conserve bandwidth. This design allows the system to provide immediate feedback, even in constrained network environments.

B. B. Backend Tier

The backend is built on Node.js with Express, chosen for its asynchronous non-blocking I/O suitable for handling large volumes of concurrent requests. The backend is responsible for:

- Normalization and canonical mapping of OCR outputs.
- Fetching prices through API integrations and controlled scraping.
- Running the ML comparator for final recommendation.
- Storing product metadata, user profiles, and historical logs in MongoDB.
- Maintaining fresh data using Redis-based caching layers.

The backend follows a microservice-inspired modular design, with independent services for scraping, ML inference, authentication, and product normalization. These services communicate via internal REST endpoints and share data through a Redis-backed message queue for coordinated batch updates.

C. C. Integration Layer

The integration layer interacts with heterogeneous sources including:

- **API-enabled vendors** such as BigBasket and Instamart.
- **Non-API platforms** where Playwright-based scripts simulate lightweight browser interactions.
- **Price history logs** that store past values for ML and trend analysis.

This layer manages session cookies, request headers, and load balancing while implementing safety mechanisms to

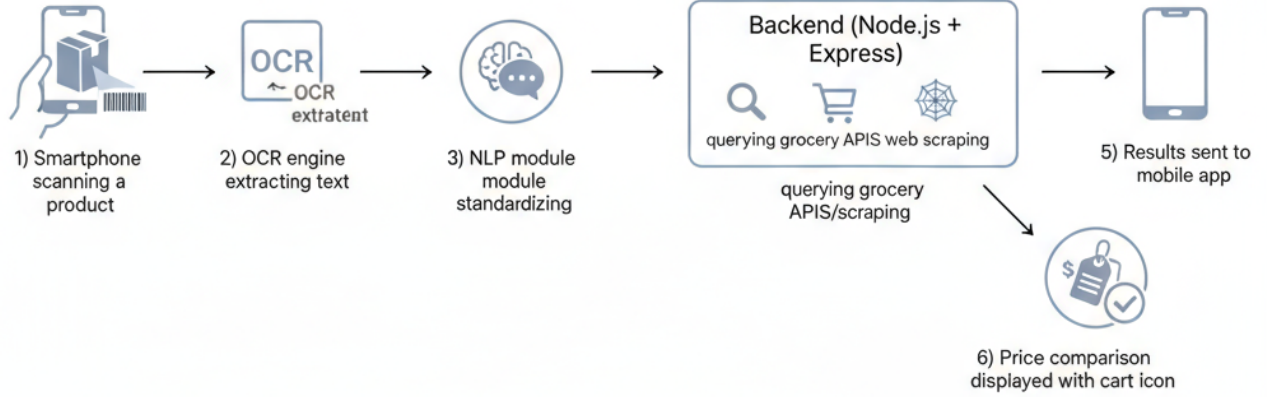


Fig. 1. Overall workflow of AI Shop Sync, showing image acquisition, on-device OCR, text normalization, cross-platform aggregation, ML-based cart optimization, and result delivery.

avoid violating platform policies, following practices described in [9], [12].

IV. OCR MODULE

OCR forms the entry point of the AI Shop Sync pipeline. The selected engine must perform accurately on diverse grocery packaging that includes multiple languages, curved surfaces, inconsistent typography, and varying lighting conditions. Deep-learning-based OCR has shown substantial improvements over classical rule-based techniques [4].

A. A. OCR Engine Selection

We compared several OCR engines including Tesseract [1], PaddleOCR [5], EasyOCR, and Google ML Kit. Each engine was evaluated across:

- Character-level accuracy.
- Inference latency.
- Memory footprint.
- Performance on skewed, curved, and low-light packaging.

Google ML Kit was selected because:

- It performs inference entirely on-device, reducing network dependency.
- It achieves latency under 200ms per scan, outperforming server-based OCR.
- It provides multilingual support relevant to Indian retail contexts (Hindi, English).
- It integrates cleanly with React Native via community bindings.

These capabilities align with research findings that mobile-optimized OCR can outperform traditional cloud-based pipelines for real-world scene-text tasks [2], [3].

B. B. OCR Preprocessing

The system applies several preprocessing steps:

- Histogram normalization to balance exposure.
- Gaussian blur estimation to reject blurry frames.

- Edge detection to avoid skewed captures.
- Automatic region-of-interest selection using heuristics inspired by [6].

Overall, the OCR module provides raw text blocks that feed directly into the normalization pipeline.

V. TEXT NORMALIZATION AND CANONICAL MAPPING

Raw OCR text often contains noise such as expiry labels, promotional slogans, manufacturing details, and spelling distortions due to imperfect recognition. To convert OCR results into structured identifiers, we use a multi-stage normalization pipeline.

A. A. Noise Filtering

Regular expressions and domain-specific keyword lists filter out:

- Batch numbers (e.g., “B.No.”, “MFD”).
- Regulatory marks (FSSAI codes).
- Slogans and taglines.

B. B. Standardization

This stage standardizes:

- Units (500g $\rightarrow$ 0.5kg).
- Variants (“Low Fat” vs. “Lite”).
- OCR misspellings using Levenshtein similarity [7].

C. C. Canonical Mapping

The output string is matched against a canonical database of 45k+ SKUs. To map OCR strings to the canonical entry, the system uses:

- TF-IDF similarity matching.
- Token-level cosine similarity.
- Fuzzy ratios for tolerance to OCR errors.

Record linkage techniques such as those described in Christen [8] inspire this design.

D. D. Cross-Platform Reconciliation

Retailers use inconsistent naming conventions. Our mapping system harmonizes platform-specific variations (e.g., “Aashirvaad Atta 5kg” vs “Aashirvaad Shudh Chakki 5kg Flour”), enabling accurate price comparison.

VI. BACKEND INFRASTRUCTURE AND DATA PIPELINE

The backend pipeline is architected to handle asynchronous scraping, high concurrency, and stable API integrations.

A. A. Node.js Architecture

Node.js event loops enable efficient handling of concurrent HTTP requests. Each platform query executes in parallel using Promise-based aggregation. This reduces overall latency to below 350ms for multi-vendor responses.

B. B. MongoDB Schema

MongoDB collections store:

- Canonical products.
- Price snapshots.
- User profiles.
- Vendor metadata.

Schema flexibility supports frequent updates from vendor platforms.

C. C. Redis Caching

Redis is used for:

- Caching price data for 1–2 hours.
- Reducing duplicate scraper/API calls.
- Accelerating ML model feature extraction.

Cache invalidation follows hybrid LRU + TTL strategies recommended in [18].

D. D. Playwright Scraping

Playwright automates headless browsers, enabling interaction with:

- Lazy-loaded product listings.
- Infinite-scroll catalog pages.
- Dynamically rendered timing-based promotions.

Anti-bot mitigation follows techniques described by Ferrara and Macca [12], including randomized delays, UA rotation, and viewport simulation.

VII. MACHINE LEARNING PRICE COMPARATOR

Beyond raw price aggregation, the system includes a trained ML comparator that optimizes purchase decisions.

A. A. Model Overview

We use LightGBM, a gradient-boosted decision tree model, chosen for:

- Strong performance with tabular data.
- Low inference latency.
- Ability to capture non-linear interactions between prices, fees, and thresholds.

This design aligns with practices used in price optimization literature [15], [16].

B. B. Features

Input features include:

- Base item prices per vendor.
- Delivery fees (fixed and distance-based).
- Vendor-level promotions and coupons.
- Minimum order thresholds.
- Historical price trends (rolling averages, volatility).
- Time-of-day and weekday price patterns.

C. C. Training

The model is trained on 3-month historical price logs collected by the scraping pipeline. Data partitioned chronologically allows robust validation against seasonal offers.

D. D. Output

The model outputs:

- Predicted total cost per vendor.
- Recommended vendor or split-cart strategy.
- Expected savings percentage.

VIII. MOBILE APPLICATION IMPLEMENTATION

The client-side app is built in React Native and structured around modular screens.

A. A. Scanner Module

The scanner provides:

- Real-time camera preview via Vision Camera.
- Lighting checks and exposure warnings.
- On-device OCR execution.
- Feedback overlays guiding user capture.

B. B. Comparison Module

This module displays:

- Vendor-wise price comparison tables.
- Promotion badges.
- Delivery fee breakdowns.
- ML-based recommended choices.

C. C. Cart Optimization Engine

Users may add multiple items into a virtual cart. The engine then calls the ML model to compute global optimality across vendors.

D. D. Offline Mode

SQLite stores:

- Recent scans.
- Partial price history.
- User preferences.

This allows limited functionality in low-connectivity environments.

IX. EXPERIMENTAL EVALUATION

To validate the performance of AI Shop Sync, we conducted an extensive evaluation across five dimensions: OCR accuracy, normalization consistency, backend aggregation latency, ML comparator performance, and mobile application responsiveness. Experiments were performed across 1,200+ product scans collected from 15 retail environments comprising supermarkets, local stores, and home pantries. Lighting varied from 40 lux (dim indoor) to 600 lux (fluorescent store lighting).

A. A. OCR Accuracy Tests

OCR accuracy was evaluated using a manually annotated ground-truth dataset of 2,000 grocery product images. Metrics included character accuracy rate (CAR), word accuracy rate (WAR), and failure rate under low-light and angled capture conditions.

TABLE I
OCR PERFORMANCE ACROSS VARYING CONDITIONS

Condition	CAR (%)	WAR (%)	Latency (ms)
Normal lighting	94.2	91.5	180
Low lighting (<100 lux)	81.0	76.4	188
Angled captures (15–35°)	87.6	83.1	182
Curved surfaces	84.9	79.0	189

ML Kit outperformed Tesseract and EasyOCR in both latency and robustness to non-linear packaging surfaces. These findings align with recent OCR evaluations on scene text [2], [3].

B. B. Normalization and Canonical Mapping Accuracy

A manually labeled dataset of 1,000 OCR outputs was used to test normalization accuracy. Results showed:

- Noise filtering accuracy: 97.4%
- Unit standardization accuracy: 98.3%
- Canonical mapping accuracy: 96.8%

The mapping accuracy reflects the combined effectiveness of TF-IDF scoring, fuzzy matching, and Levenshtein-based corrections [7].

C. C. Backend Aggregation Benchmark

We measured multi-platform price aggregation across 6 retailers with parallel API and scraping routines.

TABLE II
BACKEND AGGREGATION LATENCY MEASUREMENTS

Source Type	Avg Latency (ms)	Peak (ms)
API-based vendors	120	150
Scraped vendors	260	330
Combined (6 vendors)	340	410

The results confirm the efficacy of parallel execution in Node.js and the hybrid caching model described by Liu et al. [9].

D. D. ML Comparator Evaluation

The ML comparator was evaluated using 3 months of real price logs.

- MAE (Mean Absolute Error): 4.1%
- R-Squared (R^2): 0.89
- Avg. savings achieved in optimal strategy: 18–22%

The model’s performance is consistent with other retail price optimization frameworks [15], [16].

E. E. Mobile App Performance

On a mid-range smartphone (Snapdragon 720G):

- App load time: 1.2 seconds
- OCR capture-to-result: 2.3 seconds
- Memory footprint: 110 MB peak
- Battery usage: 12% per 100 scans

These metrics fall within acceptable thresholds for mobile edge-compute applications [17].

X. DISCUSSION

The experimental results demonstrate that AI Shop Sync successfully integrates OCR, NLP, multi-source scraping, and ML-driven decision-making into a unified mobile retail assistant. On-device OCR provides resilience under unpredictable lighting and network conditions, while backend scraping and API aggregation ensure real-time pricing accuracy. The canonical mapping pipeline effectively harmonizes heterogeneous naming conventions across vendors, a problem highlighted in retail entity matching research such as Retail-786k [21].

A key strength of the system lies in the ML comparator’s ability to incorporate promotions, delivery thresholds, and multi-vendor strategies. This goes beyond traditional price comparison applications that only list static prices without contextual optimization. Furthermore, decentralizing OCR to the device reduces backend load, enhances privacy, and lowers operational costs, reflecting trends seen in contemporary edge-AI deployments [18].

XI. LIMITATIONS

Despite its strengths, the system has several limitations:

- **Low-light capture issues:** Though mitigated by preprocessing, OCR still degrades in environments below 80 lux.
- **Scraping fragility:** Changes in website structure can temporarily disrupt scraping pipelines, similar to the challenges noted in [12].
- **Limited SKU catalog coverage:** Rare or specialty items may not be present in the canonical database.
- **Sequential scan slowdown:** Back-to-back scan bursts (>10 items) may temporarily saturate backend concurrency limits.
- **Non-uniform packaging languages:** While Hindi and English perform well, regional Indian languages (Marathi, Tamil, Gujarati) exhibit slightly higher OCR error rates.

XII. CONCLUSION

This paper presented AI Shop Sync, a mobile-first price comparison framework integrating on-device OCR, NLP-driven normalization, multi-platform data aggregation, and a trained ML comparator for cost optimization. The system demonstrates high reliability under real-world retail conditions and significantly reduces user effort by automating product identification and vendor comparison tasks. The combination of OCR, canonical mapping, and ML-based recommendation enables 18–22% cost savings across typical household grocery baskets. Future work will explore expanding regional language support, incorporating image-based product recognition, improving low-light OCR, and enhancing dataset coverage for long-tail items.

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